

The impact of meals on a probiotic during transit through a model of the human upper gastrointestinal tract.

Commercial literature on various probiotic products suggests that they can be taken before meals, during meals or after meals or even without meals. This has led to serious confusion for the industry and the consumer. The objective of our study was to examine the impact of the time of administration with respect to mealtime and the impact of the buffering capacity of the food on the survival of probiotic microbes during gastrointestinal transit. We used an in vitro Digestive System (IViDiS) model of the upper gastrointestinal tract to examine the survival of a commercial multi-strain probiotic, ProtecFlor®. This product, in a capsule form, contains four different microbes: two lactobacilli (*Lactobacillus helveticus* R0052 and *Lactobacillus rhamnosus* R0011), *Bifidobacterium longum* R0175 and *Saccharomyces cerevisiae* boulardii. Enumeration during and after transit of the stomach and duodenal models showed that survival of all the bacteria in the product was best when given with a meal or 30 minutes before a meal (cooked oatmeal with milk). Probiotics given 30 minutes after the meal did not survive in high numbers. Survival in milk with 1% milk fat and oatmeal-milk gruel were significantly better than apple juice or spring water. *S. boulardii* was not affected by time of meal or the buffering capacity of the meal. The protein content of the meal was probably not as important for the survival of the bacteria as the fat content. We conclude that ideally, non-enteric coated bacterial probiotic products should be taken with or just prior to a meal containing some fats.